

Intermediate Git — Summary

Branching Strategy

- `git switch -c feature/<name>` — create feature branches, never commit directly to `main`
- GitHub Flow: branch → push → Pull Request → review → merge → delete branch

Merging vs Rebasing

- `git merge <branch>` — joins histories, creates a merge commit, preserves the full story
- `git rebase <branch>` — replays your commits on top of another branch, produces clean linear history
- Rule: never rebase commits already pushed to a shared remote

Merge Conflicts

- Edit the conflict markers (`<<<<<<<` , `=====` , `>>>>>>>`) manually, keep what's correct
- `git add <file>` to mark resolved, then `git commit` to finalize

Pull Requests

- Push your branch to GitHub, open a PR proposing to merge into `master` / `main`
- PR shows: conversation, commits, files changed with line-by-line diff
- Merge options: merge commit / squash and merge / rebase and merge
- After merging: `git pull origin master` locally, `git remote prune origin` to clean stale refs



.gitignore

- `echo "pattern" >> .gitignore` — ignore files by name, extension, or folder
- Common ignores: `.env` , `node_modules/` , `dist/` , `*.log` , `Thumbs.db`
- Only works on untracked files — use `git rm --cached <file>` to untrack an already-committed file

git stash

- `git stash push -u -m "description"` — stash all changes including untracked files
- `git stash list` — see all stashes
- `git stash pop` — restore most recent stash and remove it from stack
- `git stash drop stash@{n}` — delete a specific stash

git cherry-pick

- `git cherry-pick <hash>` — copy a single commit from any branch onto your current branch
- The copied commit gets a new hash; the original remains where it was

reset vs revert

Command	Commits	Changes	Use when
<code>git revert <hash></code>	Adds new undo commit	Preserved	Commit is already pushed/shared
<code>git reset --soft <hash></code>	Removed	Kept staged	Local only, want to recommit
<code>git reset --mixed <hash></code>	Removed	Kept unstaged	Local only, want to re-review
<code>git reset --hard <hash></code>	Removed	Permanently deleted	Local only, discard everything